Joshua Markle

(757) 784-3838
joshuamarkle25@gmail.com
github.com/JoshuaMarkle

EDUCATION

University of Virginia
B.S. Computer Science | GPA: 3.95 *2024 - Expected 2027*

- 1st Place ICPC 2024 Mid-Atlantic Div. 2
- ICPC 2025 Mid-Atlantic Div. 1
- UVA Triathlon Team
- Baking Club

SKILLS

Languages: Python, C/C++, Bash, JavaScript, HTML/CSS, SQL, Dart
Frameworks & Tools: React, Node.js, Next.js, Supabase, Git, Linux (Arch), NeoVim, Tmux
Interests: Surfing, Snowboarding, Triathlon, Pizza Making, Reading (Tech & Nonfiction)

EXPERIENCE

UVA Career Center Chatbot *2025*

- Design, built, and presented an AI chatbot to answer student career questions using Gemini
- Implemented a RAG pipeline to provide context-aware responses

Virginia Discovery Museum Volunteer System *2025*

- Led a 4-person team in the construction of a volunteer management system using Firebase

Technical Leadership *2023*

- Taught data structures, Git, and competitive programming to 30+ students
- Organized and administered programming contests using DOMJudge on AWS
- Founded a game development club and ran 10+ hands-on workshops

PROJECTS

AlgoType | *algotype.net* *2025*

- Designed and built a public, syntax-aware code typing platform for programmers
- Implemented gamemodes, user accounts, progress tracking, payment, and authentication
- Designed a code tokenization and typing engine with real-time WPM, accuracy, and error tracking that works with 7,000+ curated code samples all within Next.js & Supabase stack

Milestone Mobile App | *play.google.com* *2023*

- Created a mobile app to help student drivers track supervised driving hours
- National winner of the Congressional App Challenge and presented to Congress

RESEARCH

Keyboard Layout Optimization Research & Mentorship *2023*

- Built a C++ system to evaluate and optimize keyboard layouts using machine learning
- Visualized layout performance using OpenGL ImGUI heatmaps and typing simulations
- Achieved over 200% efficiency improvement compared to QWERTY on modeled workloads

AWARDS

Winner of the Congressional App Challenge *2023*

- Selected as one of 374 national winners out of 11,334 participants

Xerox Award for Innovation and Information Technology *2023*

- Awarded for notable achievements in emerging technology and $20,000 scholarship

Outstanding Research Award | *NHREC Science Fair* *2023*